ETH zürich

Literature Review

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Overview of literature review

- Literature review of 13 papers
- Main topics: crack propagation in snow, snow modeling, modeling in more general sense (introduction to different modeling methods used)
- Modeling methods: Material Point Method (MPM) and Discrete Element Method (DEM)

Motivation

- Understanding the different modeling methods and their limitation is important for thesis to understand which methods are suited for what
- Understanding how cracks propagate and which factors influence them is important for risk assessment
- Understanding which factors lead to supershear crack propagation is also important for risk assessment

Modeling Methods

- Two main methods used: MPM and DEM
- DEM described by Cundall and Strack, 1979, material of disks & spheres, contact forces and resulting displacements calculated
- MPM where material is discretized in points with Eulerian background mesh to calculate stresses at points and time-dependent equations, for example in Gaume et al., 2018
- MPM is more favorable for slope-scale simulations because of modelling of grain contacts
- Other models for snow such as SNOWPACK (Bartelt & Lehning, 2002) and CROCUS (Brun et al., 1992) model snowpack response to external factors

Avalanche Release - General Crack Propagation

- All papers agree on avalanche formation by loading - weak layer failure
- Some papers discuss anticrack propagation (closure of cracks formed) like eg Gaume et al., 2018
- Others state shear cracks to be the main failure mode
- Bergfeld et al., 2025 combine the two main hypotheses: anticrack early in avalanche propagation, shearing later
- Critical crack length: length of the crack needed for breakoff (Gaume et al., 2017)

Factors influencing crack propagation

- Slope angle: higher slope angle generally increases crack speed and can trigger transition from sub-Rayleigh to supershear; it can also reduce the critical crack length needed for release (Bobillier et al., 2024; Gaume et al., 2017; Trottet et al., 2022)
- Slab stiffness (elastic modulus): stiffer slabs lead to lower normalized crack speed in some DEM results, but it tends to increase critical crack length (more crack growth needed before release) (Bobillier et al., 2024; Gaume et al., 2017)
- Weak layer strength & elastic modulus: higher weak-layer elastic modulus can increase crack speed; stronger weak layers generally require more energy/longer cracks for release (Bobillier et al., 2024; Gaume et al., 2015; Gaume et al., 2017)

Factors influencing crack propagation (continued)

- Slab density: higher slab density is associated with faster propagation in some models, while also reducing critical crack length for release (Gaume et al., 2015; Gaume et al., 2017)
- Slab thickness: thicker slabs can increase propagation speed in DEM studies; in critical-crack formulations, greater slab thickness lowers critical crack length (Gaume et al., 2015; Gaume et al., 2017)
- Weak layer thickness: thinner weak layers can lead to faster crack propagation and lower critical crack length for release (Gaume et al., 2015; Gaume et al., 2017)
- Snow structure and loading rate: sintering and lower loading rates can lead to more stable snowpacks, which means higher critical crack lenths (Mulak & Gaume, 2019)
- Loading rate: slow loading (like from snow accumulation) can lead to formation of new bonds (sintering): strengthens snowpack (Mulak & Gaume, 2019)

Deformation stages

- After loading, multiple stages of deformation
- loading response by an elastic regime, a drop in pressure and shear stress, volumetric collapse of the weak layer and lastly dense packing (Gaume et al., 2018)
- Slab bending induced by weak layer collapse at bottom of snowpack (Bergfeld et al., 2025)
- Deformation because of slab tension at top of snow (Bergfeld et al., 2025)

Crack propagation direction & mode

- Direction of crack propagation dependent on simulation scale (Trottet et al., 2022)
- Small-scale: propagation upslope (remote triggering), because collapse depth for bending limited by weak layer collapse (Trottet et al., 2022)
- Large-scale: downslope, because propagation speed can increase because of slab tension (Trottet et al., 2022)
- Mode I (normal to crack face), Mode II (tangential to existing faces) dependent on slope angle (**rosendahlModelingSnowSlab2020**)
- In steep slopes, more mode II (**rosendahlModelingSnowSlab2020**)

Supershear crack propagation I

- Supershear crack propagation when cracks propagate above elastic wave speed in snow (Trottet et al., 2022)
- Snow properties influence if supershear speed is reached
- High slab tensile strength: supershear cracks (Trottet et al., 2022)
- Increased slab tension in steep terrains: high speeds (possible supershear) (Trottet et al., 2022)
- Slope angle dependance: at below 30°, sub-rayleigh, at above 30°: supershear (Bobillier et al., 2024)
- Also property dependent (weak layer elastic modulus snow elastic modulus) (Bobillier et al., 2024)
- Generally: the "longer" cracks can propagate before breakoff, the higher the speed they can reach (direct failure in weak slabs, no failure in very strong slabs)
- Early stages of crack propagation are anticrack, later stages supershear (Bergfeld et al., 2025)

Understanding questions

- Just to make sure: MPM will be used in thesis?
- Snow data (like snow properties available?)

Ideas / Concept questions

- Differentiation between crack propagation regimes in DAS data? (Supershear vs Sub-Rayleigh)
- Different models based on loading rate to see difference in cracking of slow vs fast loading and to see effect of sintering
- Seperetate processes observed in DAS data? (Weak layer collapse, slab bending, slab breakoff etc)
- Different fracture stages observed? **gaumeInvestigatingReleaseFlow2019** talks about crown fracture, flank fracture etc
- Idea of link between snow properties - possible supershear /crack propagation speed to make assumptions based on time between initial crack - slab breakoff
- Relationship between crack propagation speed - release area like referenced in Bergfeld et al., 2025

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